

Christoffel Symbols from Basis Vector Derivatives (Polar Coordinates)

1. Coordinate System

We work in planar polar coordinates:

$$(r, \theta)$$

Position vector:

$$\mathbf{r}(r, \theta) = r \cos \theta \hat{\mathbf{x}} + r \sin \theta \hat{\mathbf{y}}$$

2. Coordinate Basis Vectors

The basis vectors are:

$$\mathbf{e}_r = \frac{\partial \mathbf{r}}{\partial r} = \cos \theta \hat{\mathbf{x}} + \sin \theta \hat{\mathbf{y}}$$

$$\mathbf{e}_\theta = \frac{\partial \mathbf{r}}{\partial \theta} = -r \sin \theta \hat{\mathbf{x}} + r \cos \theta \hat{\mathbf{y}}$$

3. Metric Check

$$\mathbf{e}_r \cdot \mathbf{e}_r = 1$$

$$\mathbf{e}_\theta \cdot \mathbf{e}_\theta = r^2$$

$$\mathbf{e}_r \cdot \mathbf{e}_\theta = 0$$

4. Key Relation

Christoffel symbols are defined by:

$$\frac{\partial \mathbf{e}_i}{\partial x^j} = \Gamma_{ij}^k \mathbf{e}_k$$

We compute each derivative explicitly.

5. Derivatives of Basis Vectors

(a) $\frac{\partial \mathbf{e}_r}{\partial r}$

$$\frac{\partial \mathbf{e}_r}{\partial r} = 0$$

Thus:

$$\Gamma_{rr}^r = 0, \quad \Gamma_{rr}^\theta = 0$$

(b) $\frac{\partial \mathbf{e}_r}{\partial \theta}$

$$\frac{\partial \mathbf{e}_r}{\partial \theta} = -\sin \theta \hat{\mathbf{x}} + \cos \theta \hat{\mathbf{y}}$$

We rewrite in terms of $\mathbf{e}_\theta$:

$$\mathbf{e}_\theta = r(-\sin \theta \hat{\mathbf{x}} + \cos \theta \hat{\mathbf{y}})$$

So:

$$\frac{\partial \mathbf{e}_r}{\partial \theta} = \frac{1}{r} \mathbf{e}_\theta$$

Thus:

$$\Gamma_{r\theta}^r = 0, \quad \Gamma_{r\theta}^\theta = \frac{1}{r}$$

(c) $\frac{\partial \mathbf{e}_\theta}{\partial r}$

$$\frac{\partial \mathbf{e}_\theta}{\partial r} = -\sin \theta \hat{\mathbf{x}} + \cos \theta \hat{\mathbf{y}}$$

Again:

$$\frac{\partial \mathbf{e}_\theta}{\partial r} = \frac{1}{r} \mathbf{e}_\theta$$

Thus:

$$\Gamma_{\theta r}^r = 0, \quad \Gamma_{\theta r}^\theta = \frac{1}{r}$$

(d) $\frac{\partial \mathbf{e}_\theta}{\partial \theta}$

$$\frac{\partial \mathbf{e}_\theta}{\partial \theta} = -r \cos \theta \hat{\mathbf{x}} - r \sin \theta \hat{\mathbf{y}}$$

Factor:

$$= -r(\cos \theta \hat{\mathbf{x}} + \sin \theta \hat{\mathbf{y}}) = -r \mathbf{e}_r$$

Thus:

$$\Gamma_{\theta\theta}^r = -r, \quad \Gamma_{\theta\theta}^\theta = 0$$

6. Final Result

The nonzero Christoffel symbols are:

$$\Gamma_{\theta\theta}^r = -r$$

$$\Gamma_{r\theta}^\theta = \Gamma_{\theta r}^\theta = \frac{1}{r}$$

All others vanish.